23BCS066

B.Tech. (Computer Engg.) V Semester Examinations

Session 2025-26

Time: 1 hr. **Computer Networks (CSC - 503) Sessional-I** **Max.Marks: 12**

Note: Attempt All Questions

1. CO1	(a) If the Transport layer already ensures reliable data transfer, what additional role does the Session layer play in maintaining communication?	2+2 (BL2, BL4)
	(b) If the Network layer fails to properly route packets but the Transport layer's reliability mechanisms are intact, explain what problems the overall communication might face.	
2. CO1	Compare and contrast physical address, IP address, and port address with respect to their scope, purpose, and layer in the OSI model. If MAC addresses are unique, explain why two devices in different parts of the world with different MAC addresses still require IP addresses to communicate.	4 (BL1, BL4)
3. CO4	Suppose a company is assigned the **192.168.16.0/22** network for its internal use. Divide this network into 6 subnets of equal size. Find the subnet ID bits, broadcast addresses, range of valid host IP addresses. Also, determine the subnet mask for these subnets.	4 (BL5)